

MINIMUM VIABLE SCALE AND FIRST-PASSAGE BELLMAN BARRIERS

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Nonconcave growth can support capital above its physical replacement threshold even when liquidation is optimal. We characterize the minimum viable scale: the unique initial productive scale at which the planner switches from extinction to growth. Separate low and high certificates, joined by local order only on the unresolved bracket, yield a four-region classification and a canonical attained first-move Bellman gap whose unique zero is the frontier. For high-state selection we introduce a first-passage Bellman-barrier certificate to nonconcave optimal growth. It partitions all histories by their first exit, combines a stay-class supersolution with an independent continuation upper bound and attained reach-and-remain policy floors, and proves exact all-optimizer selection without solving the unrestricted value function or imposing global order. Ordered, degree-5/4, and annual realizations provide exact localization, route separation, and a globally non-supermodular application.

KEYWORDS: minimum viable scale, nonconcave growth, Skiba point, Bellman barrier, first passage, exact certification, JEL: C61, E22, O41.

1. INTRODUCTION

When is productive activity large enough that growth is better than liquidation? In a non-concave economy, physical feasibility does not answer that question. A stock can reproduce itself and yet be too small to justify protecting resources for the future. We call the initial productive scale at which the planner switches from extinction to growth the *minimum vi-*

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able scale. The paper characterizes that scale and develops a method for proving global regime selection when the usual order structure is unavailable.

Four objects must be kept separate. The *physical replacement boundary* κ is the smallest positive state at which maximal current resources can replace the capital stock. The *selection frontier* K_S is the welfare threshold: below it every optimum becomes extinct, above it every optimum selects the certified high regime, and at it both regimes are optimal. The set $H = [K_{hi}, \overline{K}]$ is a proof target used to certify high selection. A stationary destination K^* , when one exists under additional conditions, is a long-run state inside the selected regime. In general,

$$0 < \kappa < K_S < K_{hi} \leq \overline{K},$$

and none of these inequalities licenses the identification of one object with another.

The resulting classification has four economically distinct regions. States below κ cannot replace their productive stock and physically contract. States between κ and K_S are physically viable but every optimal plan nevertheless liquidates. At K_S , an extinction plan and a plan selecting the high regime coexist. Above K_S , every optimizer reaches a certified high set in finite time and remains there. A destination conclusion—convergence to a particular K^* —is deliberately not part of this classification.

The threshold itself is classical. [Skiba \(1978\)](#) established indifference between distinct optimal regimes in a convex–concave growth problem, and [Dechert and Nishimura \(1983\)](#) supplied the canonical critical-capital classification. Later work studied optimal-path comparative dynamics and the behavior of critical capital as patience changes ([Amir et al., 1991](#), [Akao et al., 2011, 2025](#)). We therefore make no claim to have discovered the Skiba state, the distinction between physical and welfare thresholds, or discount-factor monotonicity.

The paper makes two contributions.

First, it *solves for the minimum viable scale* in a form that is both structural and constructive. Separate low and high certificates are joined by order only on the unresolved low-state bracket. This modular construction yields existence and uniqueness of K_S , optimizer exhaustion, extinction below the frontier, coexistence at it, all-optimizer high selection above it, strict first-move branching, target-set invariance, and an attained directional first-move Bellman gap whose unique zero is K_S . In the ordered realization the frontier has an exact rational bracket and can be localized to arbitrary prescribed precision by finite certified

value envelopes. This is more than naming a critical state: it identifies the primitive and certified obligations needed to locate it and makes clear which result owns each economic conclusion.

Second, we bring *first-passage Bellman barriers* to the nonconcave-growth problem. The method is designed for the difficult high-side question: can any complete history outside a target regime beat an explicitly feasible history inside it? It organizes all histories by their first boundary event, bounds infinite residence in the unresolved region with a residual supersolution, prices unrestricted continuation after exit with an independent Bellman upper bound, supplies attained lower witnesses from reach-and-remain policies, and compares continuations by splicing at the same state. The result is an exact, global, all-optimizer selection statement without a solution of the unrestricted value function and without a global increasing-differences assumption.

The genealogy is intentionally explicit. The first-boundary-event organization comes from first-passage analysis (Redner, 2001). Barrier-option valuation supplies the intuition that a path-dependent boundary event is the right valuation interface (Merton, 1973). Exit-time dynamic programming, Bellman inequalities, safety barriers, and primal–dual value bounds supply established ingredients (Krylov, 1982, Malisoff, 2001, Prajna and Jadbabaie, 2004, Rogers, 2002, Haugh and Kogan, 2004, Wang et al., 2015). Our contribution is their specific redesign and composition for endogenous discounted welfare: a stay-class residual bound, an independently certified unrestricted continuation bound, attained reach-and-remain witnesses, same-state first-exit splicing, and finite exact transition coverage. To our knowledge, this complete composition has not previously been used to prove all-optimizer regime selection in nonconcave optimal growth. The claim concerns the composition and application, not any borrowed ingredient or the earlier connection between first passage and option pricing.

Why is a new certificate useful? Euler equations and maximum-principle conditions identify local candidates. Phase diagrams organize stationary and often monotone motions. A computed path proves that one attractive history exists. Ordinary value iteration is valuable for discovery. None of these objects, by itself, ranks every feasible lifetime history in a nonconcave problem. A rival can decline, delay, stagnate, recover, cycle, enter and later leave, or cross a boundary more than once. The first-passage partition covers these histories

exhaustively. The proof is global in path space but economical in what it asks us to certify: class-specific upper bounds need only fall below values attained by target-class witnesses.

The method does not claim universal superiority. When global order holds, an order proof can be shorter. Validated dynamic programming, occupation measures, Hamilton–Jacobi methods, or a model-specific global argument may establish the same conclusion. The certificate is sufficient, so failure to construct one is inconclusive. It selects a regime without necessarily identifying the optimal policy within it, locating K_S , or proving convergence to K^* . The verified application is deterministic and one-state; exact coverage may become expensive in higher dimensions. These limits define the method’s proper domain rather than weakening its all-path conclusion inside that domain.

Three realizations separate the roles of theory, method, and audit. The ordered CES–log economy verifies the complete classification, destination under its extra interface, exact localization, and the patience theorem. A degree-5/4 CES economy directly realizes the strict first-passage certificate. It contains a low stationary candidate that blocks both order-based high modules, while a distinct high candidate blocks the global module’s uniqueness premise. The Bellman barrier is therefore indispensable among the paper’s three displayed high-certification modules for that economy; we do not claim that actual global order has been disproved or that no fourth proof exists. Finally, an annual economy is locally ordered on the bracket that matters but fails global increasing differences at a remote exact witness. Its classification is analytical; a finite Bellman ledger is retained only as an independent exact audit. These are theoretical implementations, not calibrations or empirical estimates.

The paper’s scope is a deterministic, one-state planner problem with endogenous labor and a compact state domain. The general result permits alternative high-certificate routes and uses local order only where the low and high anchors leave the decision unresolved. It does not establish decentralized implementation, strategic equilibrium selection, stochastic entry probabilities, or multidimensional scalability.

Section 2 defines the planner problem and states the minimum-viable-scale classification. Section 3 shows why physical viability need not imply optimal preservation. Section 4 develops and proves the first-passage Bellman barrier. Sections 5 and 6 give the ordered, patience, and annual implementations. Proofs and supporting derivations are organized by logical dependency in the Supplemental Appendix; exact certificates, audits, and the verification manifest are in the repository research appendix.

2. ENVIRONMENT AND MINIMUM VIABLE SCALE

The central object is a selection threshold, not a technological survival threshold. We therefore distinguish four objects from the outset. The lower replacement boundary κ is the smallest positive stock at which maximal current resources can replace the stock. The minimum viable scale K_S is the state at which optimal welfare switches between extinction and growth. The interval H is a certified high target used to prove selection. Finally, K^* is a destination only under additional conditions. In general,

$$\kappa < K_S < \inf H,$$

and none of these inequalities makes the objects interchangeable.

2.1. Planner's problem and Bellman objects

Let ξ index an economy. The planner solves

$$\sup_{\{C_t, L_t, K_{t+1}\}_{t \geq 0}} \sum_{t=0}^{\infty} \beta_{\xi}^t U_{\xi}(C_t, L_t) \quad (2.1)$$

subject to

$$C_t + K_{t+1} \leq \mathcal{R}_{\xi}(K_t, L_t) := (1 - \delta_{\xi})K_t + F_{\xi}(K_t, L_t), \quad (2.2)$$

where $K_t \in X = [0, \bar{K}]$, $L_t \in [0, \bar{L}_{\xi}]$, and admissible paths have $C_t > 0$ and $K_{t+1} > 0$ at every date. Define

$$\Phi_{\xi}(K) = \max_{0 \leq L \leq \bar{L}_{\xi}} \mathcal{R}_{\xi}(K, L), \quad (2.3)$$

$$r_{\xi}(K, K') = \max_{\substack{0 \leq L \leq \bar{L}_{\xi} \\ \mathcal{R}_{\xi}(K, L) > K'}} U_{\xi}(\mathcal{R}_{\xi}(K, L) - K', L), \quad (2.4)$$

with $r_{\xi} = -\infty$ when the constraint set is empty. For the formal blocks below, fix ξ and suppress its index. Put

$$\Gamma(K) = \{K' \in X : \mathcal{R}(K, L) \geq K' \text{ for some } L \in [0, \bar{L}]\},$$

$$u(K, L, K') = \begin{cases} U(\mathcal{R}(K, L) - K', L), & \mathcal{R}(K, L) > K', \\ -\infty, & \mathcal{R}(K, L) \leq K', \end{cases}$$

$$\bar{u}(c) = \sup_{0 \leq L \leq \bar{L}} U(c, L).$$

The value function is V , the optimal-successor correspondence is

$$G(K) = \arg \max_{K' \in \Gamma(K)} \{r(K, K') + \beta V(K')\},$$

and its extreme selectors are $g^-(K) = \min G(K)$ and $g^+(K) = \max G(K)$. An *extinction path* satisfies $K_t \rightarrow 0$. Given $H = [K_{\text{hi}}, \bar{K}]$, a path is *high-admissible* if it enters H in finite time and remains there thereafter.

For compactness we allow boundary actions, extend utility by $-\infty$ at nonpositive consumption, and set $V(0) = -\infty$. This closed relaxation does not change any finite-valued optimum. Under the dynamic foundations below, first-step decomposition gives

$$V(K) = \max_{\substack{0 \leq L \leq \bar{L} \\ \bar{K}' \in X}} \{u(K, L, K') + \beta V(K')\} = \max_{K' \in \Gamma(K)} \{r(K, K') + \beta V(K')\}. \quad (2.5)$$

Both maxima are attained at finite-return actions. Every use of Bellman's principle below is a feasible splice at the identical current state.

2.2. Primitive and certified blocks

Every inequality in the following blocks is required on its displayed domain.

2.2.0.1. A0 (dynamic foundations). The discount factor satisfies $\beta \in (0, 1)$; $X = [0, \bar{K}]$ and $[0, \bar{L}]$ are compact; and $\mathcal{R}$ is continuous. The correspondence Γ is nonempty, compact-valued, interval-valued, and has a closed graph; X is forward invariant and $\Gamma(0) = \{0\}$. On the closed action graph, u is upper semicontinuous, is bounded above by a finite constant M_u , and is continuous at every action at which it is finite. Utility is strictly increasing in $C > 0$, so every finite-value optimum exhausts the resource constraint and the reduced return is equivalent to the original one-period problem. From every $K > 0$ there is a feasible path with finite discounted payoff. Finally, if (K, L, K') has finite return, then for every sequence $K_n \rightarrow K$ the same pair (L, K') is feasible with positive consumption for all sufficiently large n , and $u(K_n, L, K') \rightarrow u(K, L, K')$. These conditions permit utility to be unbounded below near zero.

2.2.0.2. *A1 (low reversal).* There are $0 < \kappa < a < \bar{K}$ such that

$$\Phi(\kappa) = \kappa, \quad \Phi(K) < K \quad (0 < K < \kappa), \quad \Phi(K) > K \quad (\kappa < K \leq a).$$

There is a continuous function $h : [0, \bar{K} - \kappa] \rightarrow [0, \infty)$, with $h(0) = 0$, such that

$$\Phi(\kappa + x) - \kappa \leq h(x) \quad (0 \leq x \leq \bar{K} - \kappa).$$

Writing

$$\hat{h} = \max_{0 \leq x \leq \bar{K} - \kappa} h(x), \quad M = \sup_{0 < c \leq \hat{h}} \bar{u}(c) < \infty, \quad \Omega(z) = \sup_{0 < c \leq z} \bar{u}(c),$$

one has $\Omega(z) \rightarrow -\infty$ as $z \downarrow 0$. For every $x \in (0, a - \kappa]$, an explicitly specified feasible policy crosses below κ in finite time, has finite value $W^E(\kappa + x)$, and satisfies

$$W^E(\kappa + x) > \Omega(h(x)) + \frac{\beta M}{1 - \beta}. \quad (2.6)$$

2.2.0.3. *A2 (high-selection certificate).* There are $b < K_{\text{hi}}$, a high set $H = [K_{\text{hi}}, \bar{K}]$, a middle region $\mathcal{M} = [b, K_{\text{hi}}]$, and one of three proof routes.

In the first-passage route, bounded functions and explicit policies satisfy (C1)–(C4) and the strict-margin conditions of Theorem 3. The continuation inequality in (C4) is certified independently by the primitive supersolution and, when needed, crash-set inequalities of Lemma 4.1.

In the ordered route, successor intervals are globally nested and lower comprehensive; r has strict increasing differences on every crossed strictly feasible pair; the stationary resource and box conditions together with (2.7) have exactly one economically admissible positive candidate on X , and that candidate satisfies $K^* \in \text{int } H$. There are finite numbers $D_b < W_b$ such that an explicitly specified path from b enters H in finite time and remains there with value W_b , whereas every nonincreasing feasible path from b has value at most D_b .

In the locally ordered analytical route, there is a cutoff $B > K_{\text{hi}}$ such that successor intervals are nested and lower comprehensive and r has strict increasing differences for current states through B ; the stationary resource and box conditions together with (2.7) have no economically admissible candidate with current state in $(0, K_{\text{hi}}]$; an explicit high path from

b has value strictly above an independently derived upper bound for every nonincreasing path from b ; and $\Phi(K) < B$ for every $0 < K < K_{\text{hi}}$. Finally, every feasible transition from $[B, \bar{K}]$ to below K_{hi} admits an explicitly specified finite-prefix deviation, evaluated only with primitive returns, that rejoins the proposed path at the same finite-date state with a strictly larger discounted prefix payoff.

Each route implies, rather than assumes, that every optimum from $[b, \bar{K}]$ enters H in finite time and remains there.

2.2.0.4. *A3 (local order and no stationary trap).* For $0 < K_1 < K_2 \leq b$, $\Gamma(K_1) \subseteq \Gamma(K_2)$, and every $\Gamma(K)$ is lower comprehensive. For every $x < y$ strictly feasible at both states,

$$r(K_2, y) + r(K_1, x) > r(K_2, x) + r(K_1, y).$$

For stationary exclusion, $0 < \bar{L}$; on the positive-consumption domain, U is continuously differentiable in (C, L) , relative to $L \in [0, \bar{L}]$, with $U_C(C, L) > 0$; and $\mathcal{R}$ is continuously differentiable on the relative domain $(0, \bar{K}] \times [0, \bar{L}]$. Derivatives at $K = \bar{K}$ and the labor endpoints are one-sided. At a positive stationary triple put

$$C = \mathcal{R}(K, L) - K > 0, \quad \Psi(K, L, C) = U_C(C, L)\mathcal{R}_L(K, L) + U_L(C, L).$$

The two-date residuals $H_0 = C_0 + K_1 - \mathcal{R}(K, L_0)$ and $H_1 = C_1 + K - \mathcal{R}(K_1, L_1)$, together with active state and labor bounds, satisfy linear independence because their Jacobian with respect to (C_0, C_1) is I_2 . The complete stationary necessary system is

$$\begin{aligned} \beta \mathcal{R}_K(K, L) &= 1 && \text{if } 0 < K < \bar{K}, \\ \beta \mathcal{R}_K(\bar{K}, L) &\geq 1 && \text{if } K = \bar{K}, \\ \Psi(K, L, C) &\leq 0 && \text{if } L = 0, \\ \Psi(K, L, C) &= 0 && \text{if } 0 < L < \bar{L}, \\ \Psi(K, L, C) &\geq 0 && \text{if } L = \bar{L}. \end{aligned} \tag{2.7}$$

The resource identity, box restrictions, positivity restrictions, and (2.7) have no economically admissible solution with $K \in (0, b]$. These conditions follow from feasible two-date

and inward labor variations and do not require differentiability of V . The continuation term depends only on the successor and therefore cancels from the four-point difference.

The cross-module anchor condition is explicit:

$$0 < \kappa < a < b < K_{\text{hi}} \leq \bar{K}, \quad H = [K_{\text{hi}}, \bar{K}]. \quad (2.8)$$

It aligns independently proved low and high anchors before the selector argument is invoked.

For clarity, let DF, LR, HC, LO, and EX denote the conclusions of the dynamic, low-reversal, high-certificate, local-order/no-trap, and optimizer-exhaustion modules, and let AC denote (2.8). Splitting A2 into its first-passage, global-order, and local-analytical routes gives the complete interface map

$$\begin{aligned} A0 &\Rightarrow \text{DF}, & (\text{DF}, A1) &\Rightarrow \text{LR}, & (\text{DF}, A2^{\text{FP}}) &\Rightarrow \text{HC}_{\text{FP}}, \\ (\text{DF}, A3, A2^{\text{G}}) &\Rightarrow \text{HC}_{\text{G}}, & (\text{DF}, A3, A2^{\text{L}}) &\Rightarrow \text{HC}_{\text{L}}, \\ \text{HC} &\in \{\text{HC}_{\text{FP}}, \text{HC}_{\text{G}}, \text{HC}_{\text{L}}\}, & (\text{DF}, A3) &\Rightarrow \text{LO}, \\ (\text{DF}, \text{LR}, \text{HC}, \text{LO}, \text{AC}) &\Rightarrow \text{EX}. \end{aligned} \quad (2.9)$$

These symbols name delivered conclusions, not additional assumptions. A1, A2, and A3 never operate without the dynamic-foundation output.

2.3. The minimum-viable-scale theorem

THEOREM 1—Minimum viable scale under primitive and certified conditions: *Consider (2.1)–(2.2) on $[0, \bar{K}]$. Suppose A0 through A3 hold with*

$$0 < \kappa < a < b < K_{\text{hi}} \leq \bar{K}, \quad K_{\text{lo}} = b.$$

There is a unique $K_{\text{S}} \in (a, b)$ such that every feasible path from $0 < K \leq \kappa$ converges to zero; every optimum from $\kappa < K < K_{\text{S}}$ converges to zero; every optimum from $K_{\text{S}} < K \leq \bar{K}$ enters H in finite time and remains there; and at $K = K_{\text{S}}$ an optimal extinction path and an optimal high-admissible path coexist. Their first moves may be chosen to satisfy

$$g^-(K_{\text{S}}) < K_{\text{S}} < g^+(K_{\text{S}}).$$

Every optimum belongs to one of these two regimes. If V^- and V^+ denote the suprema over extinction and high-admissible paths, respectively, then

$$V(K) = \max\{V^-(K), V^+(K)\}. \quad (2.10)$$

At the frontier both restricted values are attained; away from it, (2.10) does not assert attainment of an unused restricted supremum.

The proof has one arrow. A0 yields existence, continuity, attainment, and a closed optimal-transition graph. A1 gives a uniform low interval; one route in A2 gives a uniform high interval; A3 gives local strong order, an empty-interior branching set, and exclusion of a low stationary trap. The all-extinction states form a down-set and the all-high states an up-set. If their boundaries differed, every intervening state would branch, putting an interval inside the branching set. Compact limits at the common boundary give the two optimal regimes and strict first moves. The full proof is in Supplemental Appendix D.

The theorem yields the four-region classification:

$$\underbrace{(0, \kappa]}_{\text{physical extinction}} \quad \underbrace{(\kappa, K_S)}_{\text{viable, optimally extinct}} \quad \underbrace{\{K_S\}}_{\text{coexistence}} \quad \underbrace{(K_S, \bar{K}]}_{\text{all optima enter } H}.$$

2.4. An intrinsic solution for the frontier

The certified target is not the definition of the economic frontier.

COROLLARY 2.1—Target-set invariance: *Maintain Theorem 1. Let $T \subseteq [0, \bar{K}]$ be closed, contain H , and exclude zero. Call a path T -admissible if it enters T in finite time and remains there. The same K_S classifies T -admissibility: every optimum below it is not T -admissible, every optimum above it is T -admissible, and both types coexist at the frontier.*

PROPOSITION 2.2—Canonical first-move frontier gap: *Maintain Theorem 1. There is a compact unresolved sub-bracket*

$$I_1 = [\underline{K}_1, \bar{K}_1] \subset (a, b), \quad \underline{K}_1 < K_S < \bar{K}_1,$$

on which both directional first-move classes contain finite-value feasible paths. For $K \in I_1$, their attained values are

$$V_1^-(K) = \max_{\substack{K' \in \Gamma(K) \\ K' \leq K}} \{r(K, K') + \beta V(K')\}, \quad V_1^+(K) = \max_{\substack{K' \in \Gamma(K) \\ K' \geq K}} \{r(K, K') + \beta V(K')\}. \quad (2.11)$$

They satisfy

$$V_1^+(K) - V_1^-(K) \begin{cases} < 0, & K < K_S, \\ = 0, & K = K_S, \\ > 0, & K > K_S, \end{cases} \quad K \in I_1. \quad (2.12)$$

Thus K_S is the unique zero on I_1 of the intrinsic, attained Bellman gap $\Delta_1 = V_1^+ - V_1^-$. At the zero, both maxima equal $V(K_S)$ and are attained by strict moves $K' < K_S$ and $K' > K_S$, respectively.

This is the general sense in which the paper solves for minimum viable scale: it identifies the unique state, supplies an intrinsic equation whose only root is that state, and permits certified rational localization without assuming a closed form.

2.5. Destination, regularity, and certified localization

These conclusions use additional interfaces and are not part of Theorem 1.

2.5.0.1. A4 (destination). There is a unique positive optimal-stationary state K^* . In addition, at least one of the following holds: (a) every high-admissible optimum satisfies $|K_{t+1} - K_t| \rightarrow 0$; (b) there are continuous functions $\mathcal{L} : H \rightarrow \mathbb{R}$ and $\rho : [0, \infty) \rightarrow [0, \infty)$, with $\rho(0) = 0$ and $\rho(z) > 0$ for $z > 0$, such that every optimal transition in H satisfies

$$\mathcal{L}(K') \leq \mathcal{L}(K) - \rho(|K - K^*|);$$

or (c) successor intervals are globally nested and lower comprehensive and r has strict increasing differences on every crossed strictly feasible pair on X .

PROPOSITION 2.3—Destination: *Maintain Theorem 1 and suppose A4 holds. Every optimum above K_S converges to K^* ; every high-admissible optimum at K_S converges to K^* ; and every extinction optimum at the frontier converges to zero.*

2.5.0.2. A5 (supplementary parameter-regularity interface). On a common compact low-state bracket I , with every $K_S(\xi)$ in its relative interior, two nonempty compact-valued path correspondences $\Pi_\xi^-(K)$ and $\Pi_\xi^+(K)$ have closed graphs and finite attained values jointly continuous in (ξ, K) . Below the frontier, the minus class contains a global extinction optimum and no plus-class path converges to zero; above it, the plus class contains a global high-admissible optimum and no minus-class path is high-admissible; at the frontier, the two classes contain the respective boundary optima. In the ordered benchmark these are the nonincreasing and nondecreasing path classes. The canonical first-move construction may be used instead when its two attained values are jointly continuous on a common sub-bracket.

PROPOSITION 2.4—Frontier regularity: *Maintain Theorem 1 on a parameter family. Work on the common compact bracket I in A5. For the canonical construction additionally require $I \subseteq I_1(\xi)$ for every parameter under consideration. Let $\widehat{V}^-$ and $\widehat{V}^+$ be either the attained A5 values or the jointly continuous canonical first-move values, and define $\Delta(\xi, K) = \widehat{V}^+(\xi, K) - \widehat{V}^-(\xi, K)$. Then Δ is negative below $K_S(\xi)$, zero at it, and positive above it; the frontier is continuous. If, on I ,*

$$|\Delta(\xi, K) - \Delta(\tilde{\xi}, K)| \leq L\|\xi - \tilde{\xi}\|, \quad \Delta(\xi, K_2) - \Delta(\xi, K_1) \geq m(K_2 - K_1)$$

for $K_2 > K_1$, then

$$|K_S(\xi) - K_S(\tilde{\xi})| \leq \frac{L}{m}\|\xi - \tilde{\xi}\|. \quad (2.13)$$

An ordered parameter change that raises Δ pointwise weakly lowers the frontier, strictly so if the gap is strictly higher at the old frontier. At a continuously differentiable regular crossing,

$$D_\xi K_S(\xi) = -\frac{D_\xi \Delta(\xi, K_S(\xi))}{\partial_K \Delta(\xi, K_S(\xi))}. \quad (2.14)$$

PROPOSITION 2.5—Finite certified localization: *Maintain Theorem 1 and A5 on a computably presented compact interval I containing K_S in its relative interior. Here rational*

points in the relative interior can be effectively enumerated, and the endpoints have directed computable rational enclosures (rational endpoints suffice). Suppose finite primitive calculations produce, for every integer $n \geq 1$ and $d \in \{-, +\}$, directed envelopes

$$\underline{V}_n^d(K) \leq \widehat{V}^d(K) \leq \overline{V}_n^d(K), \quad 0 \leq \overline{V}_n^d(K) - \underline{V}_n^d(K) \leq e_n, \quad e_n \downarrow 0,$$

uniformly on I , and comparisons of finite bounds at rational states are decidable. Define

$$\underline{\Delta}_n = \underline{V}_n^+ - \overline{V}_n^-, \quad \overline{\Delta}_n = \overline{V}_n^+ - \underline{V}_n^-.$$

For every rational $\varepsilon > 0$, a finite dovetailed search returns rational states $K_n^- < K_n^+$ such that

$$\overline{\Delta}_n(K_n^-) < 0 < \underline{\Delta}_n(K_n^+), \quad K_n^+ - K_n^- < \varepsilon, \quad K_n^- < K_S < K_n^+. \quad (2.15)$$

If the strong-crossing constant $m > 0$ is available and $|\widetilde{\Delta}_n(K_0) - \Delta(K_0)| \leq q_n$, then

$$|K_0 - K_S| \leq \frac{|\widetilde{\Delta}_n(K_0)| + q_n}{m}. \quad (2.16)$$

Supplemental Appendix E proves all results in this subsection. The closed classes in A5 are used only when their attainment and separation are explicitly available. Raw classes such as “eventually enters H ” or “converges to zero” need not be closed when entry or convergence dates escape to infinity; no strict regime-value gap is inferred from them.

3. VIABILITY IS NOT SELECTION

At the lower replacement boundary, preserving the capital base and liquidating it are economically asymmetric. Preservation protects κ and consumes only from the excess resources generated above it. As the initial excess $x = K - \kappa$ vanishes, so does that flow. A declining policy can instead release part of the base. The relevant comparison is therefore between complete lifetime histories, not between one-period marginal products. This protected-resource mechanism explains why a stock can be technologically maintainable yet optimally abandoned.

Define

$$\Phi(K) = \max_{0 \leq L \leq \bar{L}} \mathcal{R}(K, L), \quad \bar{u}(c) = \sup_{0 \leq L \leq \bar{L}} U(c, L).$$

Let $V^P(K)$ be the value restricted to paths that never cross below κ , and let $V^E(K)$ be the value restricted to paths that cross below κ in finite time. These classes exhaust the feasible paths, but neither class is assumed to be closed.

LEMMA 3.1—Physical absorption below the replacement boundary: *Suppose Φ is continuous on $[0, \kappa]$, $\Phi(\kappa) = \kappa$, $\Phi(K) < K$ for $0 < K < \kappa$, and positive consumption at $K = \kappa$ forces $K_1 < \kappa$. Every feasible path from $0 < K_0 \leq \kappa$ converges to zero.*

For a path that preserves the base, write $x_t = K_t - \kappa$. The nonlinear map h in A1 yields a scalar comparison problem. Let $\mathcal{P}_{\bar{u}, h}(x)$ be the supremum of $\sum_{t \geq 0} \beta^t \bar{u}(c_t)$ over sequences satisfying

$$x_0 = x, \quad 0 \leq x_t \leq \bar{K} - \kappa, \quad c_t > 0, \quad c_t + x_{t+1} \leq h(x_t).$$

LEMMA 3.2—Protected-resource comparison: *If $h(0) = 0$ and $\Phi(\kappa + x) - \kappa \leq h(x)$, then every preserving path satisfies*

$$C_t + x_{t+1} \leq h(x_t),$$

and

$$V^P(\kappa + x) \leq \mathcal{P}_{\bar{u}, h}(x). \quad (3.1)$$

The first comparison consumption is at most $h(x)$, while every later comparison consumption is at most $\hat{h}$. Hence

$$V^P(\kappa + x) \leq \mathcal{P}_{\bar{u}, h}(x) \leq \Omega(h(x)) + \frac{\beta M}{1 - \beta}. \quad (3.2)$$

The first term on the right tends to $-\infty$ at the boundary. A finite-valued policy that crosses below κ supplies the opposing attained welfare floor.

THEOREM 2—Viability–selection reversal: *Assume A0 and A1. Every optimum from $(\kappa, a]$ crosses below κ in finite time and converges to zero, although every state in that interval is physically maintainable. Hence physical viability and optimal selection are reversed on a nonempty right neighborhood of the replacement boundary.*

The interval-wide comparison in A1 can be obtained endogenously. Let $a_0 > \kappa$ satisfy the physical, envelope, and utility conditions of A1, and suppose that, for every $x \in (0, a_0 - \kappa]$, an explicit finite-value policy crosses below κ , with value $W^E(\kappa + x)$ satisfying

$$\liminf_{x \downarrow 0} W^E(\kappa + x) > -\infty. \quad (3.3)$$

Then there is an endogenously determined $\tilde{a} \in (\kappa, a_0]$ on which the same reversal conclusion holds.

The welfare ranking is strict:

$$V^P(\kappa + x) < W^E(\kappa + x) \leq V^E(\kappa + x) \leq V(\kappa + x). \quad (3.4)$$

Every feasible path either crosses below κ or preserves the base, so (3.4) excludes every preserving optimum. Once a path crosses, Lemma 3.1 forces extinction. Physical maintainability follows from $\Phi(K) > K$: labor attaining $\Phi(K)$ permits $K_{t+1} = K$ and positive consumption $\Phi(K) - K$. Supplemental Appendix B gives the complete proof, including the uniform argument behind the endogenous endpoint $\tilde{a}$.

3.1. A quantitative width

The comparison also produces a closed-form sufficient width.

COROLLARY 3.3—Explicit reversal width: *Maintain A0 and every condition of A1 except its strict comparison inequality. Suppose $0 < x_0 \leq a - \kappa$ and, for every $x \in (0, x_0]$,*

$$\Phi(\kappa + x) > \kappa + x, \quad h(x) \leq Cx^\rho, \quad W^E(\kappa + x) \geq w,$$

and suppose

$$\Omega(z) \leq a_U \log z + d \quad (0 < z \leq Cx_0^\rho),$$

where $C, \rho, a_U > 0$. Every

$$0 < x < \min \left\{ x_0, C^{-1/\rho} \exp \left(\frac{w - d - \beta M / (1 - \beta)}{a_U \rho} \right) \right\} \quad (3.5)$$

is physically maintainable but optimally extinguished.

The bound makes the economic forces transparent. A tighter surplus envelope, a stronger liquidation floor, or a sharper boundary singularity widens the set that can be certified. It is a sufficient certificate set, not a formula for the full extinction region or for K_S .

3.2. Beyond CES and separable utility

The reversal is not a peculiarity of a homogeneous CES aggregator.

PROPOSITION 3.4—Non-CES and nonseparable realization: *Let*

$$F(K, L) = AL^\gamma \frac{K^\theta}{1 + \lambda K^\theta}, \quad \theta > 1, \quad \gamma, \lambda > 0, \quad (3.6)$$

let $0 < \bar{L} < 1$, and let $q = 1 - \delta \in (0, 1)$. If

$$\frac{A\bar{L}^\gamma}{\theta} \left(\frac{\theta - 1}{\lambda} \right)^{(\theta-1)/\theta} > \delta, \quad (3.7)$$

then there are two positive replacement roots. At the lower root, the reversal conclusion holds for

$$U(C, L) = a(L) \log C + v(L)$$

whenever a and v are continuous on $[0, \bar{L}]$ and $0 < \underline{a} \leq a(L) \leq \bar{a} < \infty$. Rescue to a fixed interior high state is feasible on a possibly smaller interval.

The replacement ratio in (3.6) is single-peaked; (3.7) places its maximum above depreciation. A bounded derivative supplies a linear protected-resource envelope, and a zero-labor declining path supplies the finite liquidation floor. Full labor and a clipped increasing rule reach an interior high state in finite time. Thus the states in the reversal interval can be rescued, but every optimum rejects rescue. The detailed construction is in Supplemental Appendix B.

A one-period condition such as $\beta[(1 - \delta) + F_K(K, \bar{L})] > 1$ may also hold in the reversal interval. It is a technological return comparison, not the planner's Euler condition along an optimal path, and it does not rank the preservation and liquidation histories. Theorem 2 supplies only the low certificate used by Theorem 1; it neither states nor proves the global frontier.

4. FIRST-PASSAGE BELLMAN BARRIERS

The high-side question is global in path space: can any feasible lifetime history outside the target regime deliver more welfare than a feasible history that reaches and remains in it? A candidate high path proves existence, not all-optimizer selection. Euler equations characterize local necessary conditions. Phase diagrams organize selected monotone or stationary motions. Ordinary grid iteration is useful for discovery. None of these devices by itself excludes histories that decline, wait, recover, cycle, cross a boundary repeatedly, or enter the high set and later leave.

The first-passage Bellman barrier changes the proof target. It does not try to solve the unrestricted value function or identify the optimal policy. It places valid upper bounds on exhaustive classes of histories outside the target and compares them with attained values of explicitly feasible reach-and-remain policies. The result is global in the histories it covers but economical in the inequalities it requires.

4.1. *Genealogy and precise contribution*

The design deliberately combines established ideas. First-passage analysis organizes trajectories by the first boundary event (Redner, 2001, Krylov, 1982, Malisoff, 2001). Barrier-option pricing treats a path-dependent boundary crossing as the valuation interface (Merton, 1973); American-option duality shows the force of valid upper and lower value bounds without a closed-form value function (Rogers, 2002, Haugh and Kogan, 2004). Bellman inequalities and barrier certificates supply regional supersolution logic (Prajna and Jadbabaie, 2004, Wang et al., 2015).

We do not claim to have invented first-passage analysis, barrier options, Bellman supersolutions, barrier certificates, or primal–dual value bounds. Nor is this the first connection between first passage and option pricing: a barrier option is itself triggered by a first boundary crossing. The methodological contribution is the economic redesign and complete composition of five elements:

1. a first-passage partition of every feasible history outside the target;
2. a residual supersolution for histories that remain unresolved forever;
3. an independently certified upper bound on unrestricted continuation after a downward exit;

4. attained values of explicit reach-and-remain policies, used only as lower witnesses;
and
5. a splice at the first exit that preserves the common history and replaces the continuation from exactly the same state.

Finite exact transition coverage turns these inequalities into a continuum certificate. To our knowledge, this specific composition has not previously been used to prove all-optimizer regime selection in nonconcave optimal growth. The priority claim concerns the composition and application, not its borrowed ingredients.

4.2. Four exhaustive path classes

Fix

$$0 < K_{\text{lo}} < K_{\text{hi}} < \bar{K}, \quad \mathcal{M} = [K_{\text{lo}}, K_{\text{hi}}], \quad \mathcal{M}^- = [K_{\text{lo}}, K_{\text{hi}}), \quad H = [K_{\text{hi}}, \bar{K}].$$

On the feasible graph write

$$u(K, L, K') = U(\mathcal{R}(K, L) - K', L).$$

Let $V^H(K)$ be the supremum over high-admissible paths from K . Starting in $\mathcal{M} \cup H$, every feasible path belongs to one of four classes:

1. it starts in $\mathcal{M}^-$ and remains there forever;
2. before entering H , it first exits below K_{lo} ;
3. it enters H and later first exits below K_{hi} ; or
4. it enters H and remains there.

The first class is controlled by a middle barrier B_M . The two exit classes use an unrestricted continuation supersolution $\bar{V}$ and the attained policy floors W^M or W^H . The fourth class already belongs to the target. This partition is exhaustive because the state line is ordered and $\mathcal{M}^- = [K_{\text{lo}}, K_{\text{hi}})$.

The independence of $\bar{V}$ is essential. It must not be inferred from the desired equality $V = V^H$ or from the policy floors against which it will later be compared.

LEMMA 4.1—Independent continuation certification: *Let $\mathcal{D}$ be a forward-invariant state domain and let $\bar{V} : \mathcal{D} \rightarrow \mathbb{R}$ be bounded. If*

$$\bar{V}(K) \geq u(K, L, K') + \beta \bar{V}(K') \quad (4.1)$$

for every feasible transition with $K, K' \in \mathcal{D}$, then $\bar{V} \geq V$ on $\mathcal{D}$.

Alternatively, let $\mathcal{D}$ be a state domain and $\mathcal{C}$ an exit set with an independently proved bound $Z \geq V$ on $\mathcal{C}$. Suppose every path from $\mathcal{D}$ either stays there forever or first exits into $\mathcal{C}$. Require (4.1) on within- $\mathcal{D}$ transitions and, on every exit transition, require

$$\bar{V}(K) \geq u(K, L, K') + \beta Z(K'), \quad K \in \mathcal{D}, \quad K' \in \mathcal{C}. \quad (4.2)$$

Then $\bar{V} \geq V$ on $\mathcal{D}$. In both versions, $\bar{V}$ and every crash-set bound must be certified without using W^M , W^H , $V = V^H$, or any consequence of Theorem 3.

PROOF: For any path whose first T transitions remain in $\mathcal{D}$, iteration gives

$$\bar{V}(K_0) \geq \sum_{t=0}^{T-1} \beta^t u(K_t, L_t, K_{t+1}) + \beta^T \bar{V}(K_T).$$

Boundedness and $\beta^T \rightarrow 0$ prove the claim for paths that never exit. If τ is the first exit date, iterate through $\tau - 1$ and use (4.2) at τ :

$$\bar{V}(K_0) \geq \sum_{t=0}^{\tau} \beta^t u(K_t, L_t, K_{t+1}) + \beta^{\tau+1} Z(K_{\tau+1}).$$

Because $Z(K_{\tau+1}) \geq V(K_{\tau+1})$, the right side is at least the payoff of the given path. Taking the supremum proves the result. The stated independence condition makes this a primitive upper-bound argument rather than a circular use of the desired high-selection conclusion.

Q.E.D.

4.3. The certification theorem

THEOREM 3—First-passage Bellman barrier certification: *Suppose there are bounded functions*

$$B_M : \mathcal{M} \rightarrow \mathbb{R}, \quad \bar{V} : (0, K_{\text{hi}}) \rightarrow \mathbb{R}, \quad W^M : \mathcal{M}^- \rightarrow \mathbb{R}, \quad W^H : H \rightarrow \mathbb{R}$$

with the following properties.

(C1) The state domain $[0, \bar{K}]$ is forward invariant under every feasible transition from $\mathcal{M} \cup H$.

(C2) For every feasible transition from $K \in \mathcal{M}^-$ to $K' \in \mathcal{M}^-$,

$$B_M(K) \geq u(K, L, K') + \beta B_M(K'), \quad B_M(K) \leq W^M(K). \quad (4.3)$$

(C3) There are explicitly specified high-admissible policies from $\mathcal{M}^-$ and H , with attained values J^M and J^H , such that

$$W^M \leq J^M \quad \text{on } \mathcal{M}^-, \quad W^H \leq J^H \quad \text{on } H. \quad (4.4)$$

(C4) The inequality $\bar{V} \geq V$ below K_{hi} has been proved independently by Lemma 4.1, including any required crash-set bound, and

$$W^M(K) \geq u(K, L, K') + \beta \bar{V}(K') \quad \text{if } K \in \mathcal{M}^-, K' < K_{\text{lo}}, \quad (4.5)$$

$$W^H(K) \geq u(K, L, K') + \beta \bar{V}(K') \quad \text{if } K \in H, K' < K_{\text{hi}}. \quad (4.6)$$

Then

$$V(K) = V^H(K) \quad \text{for every } K \in \mathcal{M} \cup H. \quad (4.7)$$

If there is $\varepsilon > 0$ such that $B_M(K) + \varepsilon \leq W^M(K)$, and (4.5)–(4.6) hold with ε added to their right sides, every optimum from $\mathcal{M} \cup H$ is high-admissible. Thus the theorem supplies the high-certificate interface and the high anchor H .

PROOF: Fix a feasible path π from $K_0 \in \mathcal{M} \cup H$. We compare it with an explicit high-admissible path.

Case 1: the path starts in $\mathcal{M}^-$ and remains there forever. Iterating (C2), for every $T \geq 1$, gives

$$\sum_{t=0}^{T-1} \beta^t u_t \leq B_M(K_0) - \beta^T B_M(K_T).$$

Boundedness makes the terminal term converge to zero. Hence

$$J(\pi) \leq B_M(K_0) \leq W^M(K_0) \leq J^M(K_0) \leq V^H(K_0).$$

Case 2: before entering H , the path exits below K_{lo} . Let τ be its first such exit date. Its continuation value at $K_{\tau+1}$ is at most $\bar{V}(K_{\tau+1})$, so its value is no larger than

$$\sum_{t=0}^{\tau-1} \beta^t u_t + \beta^\tau \{u_\tau + \beta \bar{V}(K_{\tau+1})\}.$$

By (4.5), the term in braces is at most $W^M(K_\tau) \leq J^M(K_\tau)$. Follow the original history through K_τ , then use the explicit policy whose value is $J^M(K_\tau)$. The splice is feasible, begins its replacement at exactly the same state, and is high-admissible. It weakly dominates π .

Case 3: the path enters H but later leaves it. Let τ be the first date at which the current state is in H and the successor is below K_{hi} . As in Case 2, $\bar{V}$ bounds the continuation from that successor. Inequality (4.6) replaces the exit and all subsequent choices by the explicit policy whose attained value $J^H(K_\tau)$ is at least $W^H(K_\tau)$. The common prefix cancels, and the spliced high-admissible path weakly dominates π .

Case 4: none of the preceding cases occurs. If the path begins in H and never leaves, it is high-admissible. If it begins in $\mathcal{M}^-$, it either enters H and remains there or is covered by Cases 1–3. The four possibilities exhaust all feasible paths.

Every feasible path is therefore weakly dominated by a high-admissible path, so $V \leq V^H$. The reverse inequality follows because high-admissible paths are feasible, proving (4.7). Under the strict conditions, an infinite middle path loses at least ε in Case 1. A finite exit path loses at least $\beta^\tau \varepsilon > 0$ in Case 2 or Case 3. No non-high path can attain V , and every optimum is high-admissible. Notice that this is pathwise strictness, not a uniform value gap between nonclosed path classes: a very late exit may lose only a small discounted positive amount. *Q.E.D.*

4.4. Finite exact realization

The theorem is an analytical statement about the continuum. A finite ledger is one way to certify its inequalities exactly.

PROPOSITION 4.2—Finite exact realization and primitive robustness: *The first-passage theorem has the following finite realization.*

1. *If a finite log-state grid covers compact positive current-state and successor domains; barriers are affine in log capital on each cell; every unrestricted transition box has an outward rational reward upper bound; every prescribed-policy box has an outward rational reward lower bound; every such box has an outward feasibility enclosure; all relevant transition boxes are included in the recorded coverage; and every residual, continuation, policy-subsolution, sandwich, closure, handoff, and exit row is verified at the relevant endpoints with a positive exact margin, then the ledger implies (C1)–(C4) on the continuum.*
2. *If continuous barriers and finitely many continuous feedback policies satisfy a common strict gap; the policies have uniform positive-consumption, finite-entry, and strict-invariance margins; one-period rewards are uniformly bounded in absolute value along certified trajectories; all primitive objects have computable moduli of continuity; and the barriers, feedback policies, resource and reward primitives admit algorithms producing directed rational upper and lower enclosures at rational boxes and policy nodes to any requested accuracy, then a finite rational endpoint ledger with positive margins exists.*
3. *On a fixed finite direct-successor architecture with rational grid nodes, discount factor, reward bounds, and policy data, existence of a positive certificate is exactly decidable by invariant-core deletion, finite-state dynamic programming, backward induction, finite policy enumeration, and a rational linear program for each selected policy graph.*
4. *Let $m_\ell(\xi)$ index every recorded residual, continuation, policy-subsolution, sandwich, closure, handoff, exit, feasibility, consumption, entry, invariance, and coverage margin. If $\mu_\ell = m_\ell(\xi_0) > 0$, and S_ℓ is a local Lipschitz bound on a ball of radius r_0 , the same certificate remains valid whenever*

$$\|\xi - \xi_0\|_\infty < r_{\text{cert}} := \min \left\{ r_0, \min_\ell \frac{\mu_\ell}{2S_\ell} \right\}, \quad (4.8)$$

with a positive margin divided by zero sensitivity interpreted as $+\infty$.

The conclusion depends on the recorded proof object and exact verification, not on the program that proposed the grid or policy graph.

Supplemental Appendix C proves the proposition. The repository research appendix records the transition coverage, outward bounds, row families, exact margins, and

constructor–verifier trust boundary. The finite object is not a pointwise grid sample: end-point soundness and complete box coverage are what lift it to the continuum.

4.5. A mild-return CES realization

The method is not confined to a degree-two benchmark or to extreme returns to scale. Consider the rational annual economy

$$\begin{aligned} F(K, L) &= \frac{84}{125} \left(\frac{2KL}{K+L} \right)^{5/4}, \\ U(C, L) &= \frac{7}{10} \log C + \frac{3}{10} \log(1 - L), \\ (\beta, \delta, \bar{L}) &= \left(\frac{24}{25}, \frac{2}{25}, \frac{22}{25} \right), \quad X = \left[0, \frac{35}{2} \right]. \end{aligned} \tag{4.9}$$

Let κ be the smaller positive solution of $F(K, \bar{L}) = \delta K$, and put

$$\mathcal{M} = \left[\frac{1}{20}, \frac{2}{25} \right], \quad H = \left[\frac{2}{25}, \frac{35}{2} \right].$$

PROPOSITION 4.3—Degree-5/4 CES Bellman-barrier realization: *For the economy in (4.9):*

1. *The replacement boundary obeys $0 < \kappa < \frac{7}{10^6}$. Every feasible finite-return path from $0 < K_0 \leq \kappa$ converges to zero. The analytical surplus bound proves optimal extinction on*

$$\kappa < K_0 \leq \kappa + \kappa \left(\frac{46}{125} \right)^{25}.$$

2. *The recorded piecewise log-affine barriers and explicit policy floors satisfy (C1)–(C4), with a strictly positive margin in every row family. Consequently,*

$$V(K) = V^H(K) \quad \left(\frac{1}{20} \leq K \leq \frac{35}{2} \right), \tag{4.10}$$

and every optimum from this interval reaches $H = [2/25, 35/2]$ in finite time and remains there.

3. *No positive optimal-stationary state lies in $[1/40000, 1]$; consequently no optimal path converges to a state in that interval. Every positive optimal-stationary state has $s =$*

K/L equal to one of the two positive roots of

$$8D^3(1+s)^9\{D(47s+12)-15\delta\}-21875A^4s^2=0,$$

$$D:=\beta^{-1}-(1-\delta)=\frac{73}{600}, \quad A=\frac{84}{125}. \quad (4.11)$$

One root lies in $(0, 1/1000)$ and the other in $(2, 3)$; their capital stocks lie, respectively, below $1/40000$ and above one. Both roots generate interior positive stationary KKT candidates through

$$L(s)=\frac{2555s}{3431s+156}, \quad K(s)=\frac{2555s^2}{3431s+156}, \quad C(s)=\frac{511s^2(73s+13)}{150(3431s+156)}.$$

The clean exact replay checks 390,842 proof rows and 3,839 coverage cells, for 394,681 recorded rows or cells. The smallest replayed strict margin is approximately 4.45756×10^{-9} ; within the independently certified unrestricted-continuation family it is approximately 1.68755×10^{-7} . A separate exact stationary audit covers 1,959 boxes. These counts describe the retained proof object, not the method's conceptual definition or the search routine that discovered it.

The proposition is a direct realization of Theorem 3, not a second classification theorem. Its low-state and stationary conclusions record the scope of the example and do not replace the barrier.

COROLLARY 4.4—Route separation: *For the degree-5/4 economy in Proposition 4.3, neither order-based high route in A2 is available. The first-passage Bellman-barrier route is therefore indispensable among the three high-certification modules of this paper's architecture.*

Indeed, the lower stationary candidate satisfies $0 < K_{\text{low}}^{\text{cand}} < 1/40000 < 1/20 = b$. It meets the stationary resource, Euler, and labor equations with positive labor and consumption, so the no-low-stationary-candidate condition shared by the two order-based routes fails. The second candidate above one separately defeats the global route's unique-positive-candidate premise. The first-passage route nevertheless proves (4.10).

This is an architectural necessity result, not a universal proof-theoretic claim. It does not establish that actual global order itself fails in this economy, and it does not exclude a

fourth, logically different proof. The annual globally non-supermodular economy studied later is a separate application with an analytical locally ordered high proof; it is not the route-separation example.

4.6. *What the method does and does not deliver*

The method's comparative advantage is exact class-level selection when an order theorem is unavailable or unnecessarily strong. It covers every lifetime history, asks only for sign-separating bounds, and permits a small trusted verifier to audit a certificate proposed by a more elaborate search. Ordered arguments may be simpler when their premises hold; validated dynamic programming, HJB methods, occupation-measure methods, or model-specific global optimization could also prove the same conclusion.

The barrier remains a sufficient method. Failure to construct a certificate for another parameter vector is inconclusive: the regions may be poorly chosen, the policy floor weak, the continuation bound loose, the partition coarse, or high selection false. Non-certification by itself distinguishes none of these cases. The theorem classifies a target regime; it need not identify the policy within that regime, locate K_S , or imply convergence to K^* . Its verified realization is deterministic and one-state. Stochastic, strategic, and multidimensional extensions require new theorems, not extrapolation from the present certificate.

5. SCALE, LOCALIZATION, AND PATIENCE

The classification in Theorem 1 separates four objects that are often compressed into a single “critical capital” number. The physical replacement boundary κ is the lower weak-replacement boundary: exact replacement there leaves no consumption, so admissible finite-return paths must move down. The selection frontier K_S is the boundary, or infimum, of the open all-high region; at the frontier itself extinction and high selection coexist. The certified high set H is a proof target, not an economic definition. A positive stationary destination K^* is an additional asymptotic conclusion. In general, $\kappa < K_S < \inf H$; when the separate destination interface applies, the selected limit K^* lies in the closed high set. Each relation has a different proof. This section maps the first and last objects to primitives, realizes the selection theorem in an ordered CES–log economy, localizes its frontier exactly, and then asks how the frontier moves with patience.

5.1. Primitive scale and stationary objects

Assume $\theta > 1$, $0 < \delta < 1$, $\beta \in (0, 1)$, and $\bar{L} > 0$. Let production be homogeneous of degree θ ,

$$F(K, L) = L^\theta y(K/L), \quad F(K, 0) = 0, \quad (5.1)$$

with the continuous extension displayed at $L = 0$, where $y(0) = 0$ and y is continuously differentiable on the relevant positive domain. Assume

$$\bar{a} = \max_{k>0} \frac{y(k)}{k}, \quad \bar{\varphi} = \max_{k>0} y'(k), \quad 0 < \bar{a}, \bar{\varphi} < \infty.$$

Put $q = 1 - \delta$ and assume $\mu_\beta = \beta^{-1} - q > 0$. A productivity wedge τ_A reduces usable output by $1 - \tau_A$, a return wedge ω_R raises the required marginal return, and effective labor capacity is $L_c = (1 - \tau_L)\bar{L}$, where $0 \leq \tau_A, \tau_L < 1$ and $\omega_R \geq 0$.

PROPOSITION 5.1—Physical and marginal capacity floors: *Suppose maximizers k_a and k_φ of $y(k)/k$ and $y'(k)$ satisfy $k_a L_c, k_\varphi L_c \leq \bar{K}$. The labor capacities required for physical replacement and for attainment of the required marginal return are*

$$L_{\text{avg}}(\tau_A) = \left(\frac{\delta}{(1 - \tau_A)\bar{a}} \right)^{1/(\theta-1)}, \quad L_{\text{mg}}(\tau_A, \omega_R) = \left(\frac{\mu_\beta + \omega_R}{(1 - \tau_A)\bar{\varphi}} \right)^{1/(\theta-1)}. \quad (5.2)$$

If $L_c < L_{\text{avg}}$, every feasible capital path converges geometrically to zero. If $L_c > L_{\text{avg}}$, some positive stock has strict replacement surplus. The required marginal return is attainable if and only if $L_c \geq L_{\text{mg}}$. Moreover,

$$\frac{L_{\text{mg}}}{L_{\text{avg}}} = \left(\frac{(\mu_\beta + \omega_R)\bar{a}}{\delta\bar{\varphi}} \right)^{1/(\theta-1)}. \quad (5.3)$$

A common productivity wedge moves both floors proportionally and leaves their ratio unchanged.

Repository Appendix R6 gives the associated joint wedge surfaces, boundary elasticities, physical-collapse threshold, and stationary productivity signs.

The proposition distinguishes capacity to avoid physical collapse from capacity to support the intertemporal marginal return. Neither floor is the minimum viable scale: K_S compares complete lifetime programs.

For the degree-two CES scale technology used below,

$$F_A(K, L) = 4A \frac{K^2 L^2}{(K + L)^2}, \quad A > 0, \quad (5.4)$$

put $m(A) = 2A\bar{L}/\delta - 1$. The full-labor weak-replacement set is nonempty if and only if $m(A) \geq 1$. When $m(A) = 1$ it is the singleton $\{\bar{L}\}$. When $m(A) > 1$ its endpoints are

$$\kappa(A) = \bar{L}\{m(A) - \sqrt{m(A)^2 - 1}\}, \quad \bar{\kappa}_{\text{sus}}(A) = \bar{L}\{m(A) + \sqrt{m(A)^2 - 1}\}. \quad (5.5)$$

They obey

$$\kappa(A) \bar{\kappa}_{\text{sus}}(A) = \bar{L}^2, \quad \log \frac{\bar{\kappa}_{\text{sus}}(A)}{\kappa(A)} = 2 \operatorname{arccosh} m(A). \quad (5.6)$$

Strict positive-consumption maintenance is feasible only between the roots.

Stationarity can also be reduced to primitives without differentiating the value function.

Under log utility

$$U(C, L) = \eta \log C + (1 - \eta) \log(1 - L), \quad 0 < \eta < 1, \quad 0 \leq L \leq \bar{L} < 1, \quad (5.7)$$

an interior positive stationary optimum satisfies

$$F_K(K, L) = \mu_\beta, \quad \eta(1 - L)F_L(K, L) = (1 - \eta)C, \quad C = F(K, L) - \delta K. \quad (5.8)$$

The complete conditions also include the one-sided state and labor inequalities at $K = \bar{K}$, $L = 0$, and $L = \bar{L}$; they are stated and proved in Supplemental Appendix F.

Define, wherever $y'(k) > 0$,

$$\ell(k) = \left(\frac{\mu_\beta}{y'(k)} \right)^{1/(\theta-1)}$$

and let the admissible ratio set be

$$\mathcal{K} = \left\{ k > 0 : 0 < \ell(k) < \bar{L}, \quad 0 < k\ell(k) < \bar{K}, \quad \mu_\beta \frac{y(k)}{y'(k)} - \delta k > 0 \right\}.$$

On $\mathcal{K}$, put

$$\Xi(k) = \eta[1 - \ell(k)]\mu_\beta \left[\theta \frac{y(k)}{y'(k)} - k \right]$$

$$- (1 - \eta)\ell(k) \left[\mu_\beta \frac{y(k)}{y'(k)} - \delta k \right]. \quad (5.9)$$

Zeros of Ξ in $\mathcal{K}$ generate all interior candidates through

$$L = \ell(k), \quad K = k\ell(k), \quad C = \ell(k) \left[\mu_\beta \frac{y(k)}{y'(k)} - \delta k \right]. \quad (5.10)$$

If $y'' < 0$, $\{y/(ky')\}' \geq 0$, and $\theta\delta \geq \mu_\beta$ on the admissible interval, Ξ has at most one zero there. Endpoint signs give a candidate; the boundary inequalities exclude the remaining cases. This is a uniqueness statement for primitive stationary candidates. Optimal existence comes separately from a limiting optimal path and closedness of the optimal transition graph.

5.2. An ordered CES-log realization

Consider (5.4)–(5.7) on $[0, 50]$ with

$$(\beta, \delta, \eta, \bar{L}) = \left(\frac{7}{10}, \frac{7}{10}, \frac{7}{10}, \frac{22}{25} \right), \quad A \in \mathcal{A} := \left[\frac{236}{25}, \frac{1319}{125} \right], \quad H = [2, 50]. \quad (5.11)$$

The labor-reduced return has strict increasing differences on the whole state domain. An exact water-filling bound dominates every nonincreasing history from a low high-side anchor, while a rational escape ladder reaches H . The stationary reduction selects one positive optimal stationary stock above 2. The low-side comparison in Theorem 2 supplies the opposite anchor.

COROLLARY 5.2—Ordered realization of minimum viable scale: *For every $A \in \mathcal{A}$ there is a unique frontier*

$$K_S(A) \in \left(\kappa(A), \frac{11}{320} \right).$$

Every feasible path from $0 < K \leq \kappa(A)$ becomes extinct. Every optimum from $\kappa(A) < K < K_S(A)$ becomes extinct; every optimum from $K_S(A) < K \leq 50$ reaches H and converges to the unique positive stationary destination $K^(A) > 2$. At $K_S(A)$ an extinction optimum and a high optimum coexist and their first moves branch strictly.*

The destination in this corollary uses global order and unique stationarity; it is not part of the generic selection conclusion. On the low-state bracket, define the attained monotone-

path values

$$V_A^\uparrow(K) = \max_{K_{t+1} \geq K_t} \sum_{t \geq 0} \beta^t U(C_t, L_t), \quad V_A^\downarrow(K) = \max_{K_{t+1} \leq K_t} \sum_{t \geq 0} \beta^t U(C_t, L_t).$$

Their difference is negative below $K_S(A)$, zero at it, and positive above it. Thus the frontier is an attained value comparison, not a grid-defined number.

5.3. Exact and arbitrary-precision localization

The general localization result in Proposition 2.5 becomes constructive here. Finite-horizon monotone problems have explicit upper and lower terminal envelopes. Directed rational bounds on their logarithms and interval branch-and-bound give converging rational enclosures for $V_A^\uparrow$ and $V_A^\downarrow$. A dovetailed search over rational states on both sides of the zero cannot stall at a trial state that happens to equal the frontier.

PROPOSITION 5.3—Exact benchmark bracket and constructive localization: *Uniformly for $A \in \mathcal{A}$,*

$$\kappa(A) + \frac{3}{40000} < K_S(A) < \frac{31}{1000}. \quad (5.12)$$

For rational A , and for A supplied by computable directed rational enclosures, an algorithm returns a finite exact rational bracket for $K_S(A)$ of any prescribed positive width. On compact rational parameter cells the same conclusion holds whenever the reported endpoint signs have uniform strict margins.

The bracket has an economic use. Define

$$c_A = \frac{\delta \bar{K}_{\text{sus}}(A)^2}{4\bar{L}[\bar{K}_{\text{sus}}(A) + \bar{L}]}, \quad \varrho_A = \left(\frac{q}{1 + c_A} \right)^{1/(1-\beta)}, \quad x_A^* = \frac{\kappa(A)\varrho_A}{1 - \varrho_A}. \quad (5.13)$$

For $0 < \alpha < 1$, put $K_\alpha = \kappa(A) + \alpha x_A^*$. Both a replacement-preserving path and a path reaching H are feasible from this state. Nevertheless both are strictly dominated by liquidation.

COROLLARY 5.4—Selection-aware augmentation: *At the primitives (5.11),*

$$V_A(K_\alpha) - V_A^H(K_\alpha) \geq V_A(K_\alpha) - V_A^{\text{surv}}(K_\alpha) \geq \frac{7}{3} \log \left(\varrho_A + \frac{1 - \varrho_A}{\alpha} \right) > 0, \quad (5.14)$$

and the lower bound diverges as $\alpha \downarrow 0$. If the initial stock is augmented before optimization, the exact regime-changing amount is

$$\Delta_S(A, \alpha) = K_S(A) - K_\alpha, \quad (5.15)$$

with

$$\frac{3}{40000} - \alpha x_A^* < \Delta_S(A, \alpha) < \frac{31}{1000} - \kappa(A) - \alpha x_A^*. \quad (5.16)$$

Below this amount all optima become extinct; at it the two regimes coexist; above it every optimum reaches H and converges to $K^*(A)$, provided the augmented stock remains in the state domain.

This is a planner comparative experiment. It neither specifies a financing technology nor asserts a decentralized policy prescription. Its point is narrower: from a stock below replacement, $\max\{\kappa - K_0, 0\}$ is the physical-maintenance shortfall. At the states $K_\alpha > \kappa$ studied here that shortfall is already zero, whereas $K_S - K_\alpha$ is still required to change the optimally selected regime.

5.4. Patience

Hold $(\delta, \eta, \bar{L}) = (7/10, 7/10, 22/25)$ and $A \in \mathcal{A}$ fixed, but let $\beta \in [7/10, 1)$. For this exercise put

$$\varrho_{A,\beta} = \left(\frac{q}{1 + c_A} \right)^{1/(1-\beta)}.$$

Greater patience raises the value of future productive scale, but in a nonconcave problem that observation alone does not order optimizers. The ordered realization yields the stronger, pathwise statement.

A nearby benchmark is the dynamic adjustment-cost theorem of [Dekel et al. \(2025\)](#). It does not subsume the result below: their parameter shifts a period return while the discount factor stays fixed and adjustment costs enter separately. Here changing β changes every future weight and the Bellman fixed point, endogenous labor changes the resource correspondence, and the conclusion orders every optimizer path rather than one ordered selection. We claim this specialization and conclusion, not priority over dynamic monotone comparative statics.

THEOREM 4—Patience lowers minimum viable scale: *For every fixed $A \in \mathcal{A}$, the selection frontier $K_S(A, \beta)$ is strictly decreasing and continuous on $\beta \in [7/10, 1)$, while the unique positive stationary destination $K^*(A, \beta) > 2$ is strictly increasing and continuous. The frontier is jointly continuous in (A, β) and satisfies*

$$\kappa(A) + \frac{\kappa(A)\varrho_{A,\beta}}{1 - \varrho_{A,\beta}} \leq K_S(A, \beta) \leq K_S(A, 7/10) < \frac{31}{1000}, \quad (5.17)$$

and

$$\lim_{\beta \uparrow 1} K_S(A, \beta) = \kappa(A). \quad (5.18)$$

Moreover, if $7/10 \leq \beta_1 < \beta_2 < 1$, then from every common positive initial state every β_2 -optimal path lies weakly above every β_1 -optimal path at every positive date:

$$K_t^{\beta_1} \leq K_t^{\beta_2} \quad (t \geq 1).$$

The proof does not compare selected policy functions. Strict increasing differences in the discount factor strongly orders the entire optimal successor correspondences. Induction then orders every pair of optimal paths. At the old frontier, every more-patient optimizer is high, so the new frontier lies strictly below it. Finally, any fixed stock above κ has a finite rescue path to a high stationary tail whose utility advantage is multiplied by $(1 - \beta)^{-1}$; this proves (5.18). The complete argument is in Supplemental Appendix G.

6. LOCAL ORDER WITHOUT GLOBAL ORDER

The ordered benchmark is useful because it delivers a destination and global comparative statics. Global order, however, is not needed to identify minimum viable scale. This section gives an annual-frequency implementation in which strict increasing differences holds on the entire region used to resolve the selection frontier but fails elsewhere in the state space. The example is a theoretical implementation, not an empirical calibration.

Use the degree-two CES technology (5.4), log utility (5.7), and

$$\beta = \frac{24}{25}, \quad \delta = \frac{2}{25}, \quad \eta = \frac{7}{10}, \quad \bar{L} = \frac{22}{25}, \quad A \in \mathcal{A}^{\text{ann}} := \left[\frac{2479}{250}, \frac{251}{25} \right]. \quad (6.1)$$

The state domain is $[0, 450]$, the high anchor is $b = 2/125$, and $H = [3/2, 450]$.

PROPOSITION 6.1—Annual analytical anchors: *Uniformly over $A \in \mathcal{A}^{\text{ann}}$:*

1. *The lower replacement root satisfies*

$$\frac{1}{500} < \kappa(A) < \frac{2027}{10^6}. \quad (6.2)$$

Every feasible path from $0 < K \leq \kappa(A)$ converges to zero, and every optimum from

$$\kappa(A) < K \leq \kappa(A) + \frac{1}{500000}$$

crosses below $\kappa(A)$ in finite time and becomes extinct.

2. *Every optimum from $[2/125, 450]$ reaches H in finite time and never leaves it. The proof uses an exact bound on declining paths, a rational escape ladder, local strong order through capital 30, and a two-date no-exit comparison above 30.*

3. *There is a unique positive stationary candidate. Writing $s^* = K^*/L^*$,*

$$21 < s^*(A) < 22, \quad \frac{25039}{1444} < K^*(A) < \frac{494648}{27233}. \quad (6.3)$$

Every positive convergent optimal path converges to this candidate.

The last sentence is deliberately conditional on convergence. The annual argument does not prove that every high path converges.

Local order follows from a primitive calculation. For the CES–log return, strict increasing differences holds on every strictly feasible rectangle whose current states do not exceed an explicit cutoff K_{ID} . At the annual primitives $K_{\text{ID}} > 30$ uniformly. This covers the entire unresolved selection bracket and the unique stationary candidate. It does not extend to the entire state space.

PROPOSITION 6.2—Exact failure of global increasing differences: *Let r_A be the annual labor-reduced one-period return and define*

$$\mathfrak{D}_A = r_A(39, 7) + r_A(36, 4) - r_A(39, 4) - r_A(36, 7). \quad (6.4)$$

Then $\mathfrak{D}_A < 0$ for every $A \in \mathcal{A}^{\text{ann}}$, although strict increasing differences holds for current capital through 30.

The witness at current states 36 and 39 blocks any proof that assumes global increasing differences on $[0, 450]$. It lies outside both the frontier neighborhood and the stationary bracket. It therefore says nothing about whether nonorder near minimum viable scale changes the selection mechanism.

COROLLARY 6.3—Locally ordered, globally non-supermodular realization: *Uniformly for $A \in \mathcal{A}^{\text{ann}}$, there is a unique frontier*

$$\kappa(A) + \frac{1}{500000} < K_S^{\text{ann}}(A) < \frac{2}{125}. \quad (6.5)$$

Below it every optimum becomes extinct; above it every optimum reaches and remains in $H = [3/2, 450]$; at it an optimal path of each kind coexists and the first moves branch strictly.

This is an application of Theorem 1. It asserts selection and optimizer exhaustion. It does not assert a common destination for every high path. The distinction is substantive: global order is one of the separate routes that can turn high selection into convergence, and that route is unavailable here.

The analytical result is locally stable in primitives. Fix $\eta = 7/10$ and let

$$\xi_0 = \left(\frac{4989}{500}, \frac{24}{25}, \frac{2}{25}, \frac{22}{25} \right) \quad (6.6)$$

in the coordinate order $(A, \beta, \delta, \bar{L})$.

PROPOSITION 6.4—Annual primitive robustness box: *On the closed box $\|\xi - \xi_0\|_\infty \leq 10^{-8}$, the following claims hold simultaneously: the moving replacement root remains in the bracket (6.2); every optimum through $\kappa(\xi) + 1/500000$ becomes extinct; every optimum from $2/125$ reaches and remains in H ; every positive optimal-stationary state lies in $(17, 19)$; local strict increasing differences holds through current capital 30; global increasing differences fails at the four arguments in (6.4); and the unique selection frontier retains the strict bracket (6.5) and the complete extinction/coexistence/high-selection classification.*

All claims in this section are analytical. The retained annual Bellman ledger, low-side ledger, stationary checks, nonorder script, and robustness box verifier are independent exact

audits. They are not premises of Corollary 6.3. Supplemental Appendix H gives the proof; the audit map and exact row counts appear in the repository appendix.

7. CONCLUSION

Minimum viable scale is the welfare-selection frontier K_S , not the physical replacement stock κ . Below κ , extinction is a technological fact. Between κ and K_S , survival is feasible but liquidation is optimal. At K_S , extinction and growth coexist. Above it, every optimizer selects high scale. This distinction is what makes the frontier economically useful: support sufficient to preserve a stock need not be sufficient to change the selected regime.

The first contribution is to solve for this frontier at the level of all optimal histories. Separate low and high certificates, joined by local order only on the unresolved bracket, deliver optimizer exhaustion, strict first-move branching, an intrinsic attained Bellman gap, and constructive localization. The ordered realization sharpens the result to an exact rational bracket and shows that greater patience strictly lowers minimum viable scale toward the replacement boundary.

The second contribution is the first-passage Bellman-barrier method. Its stay-class super-solution, independently bounded continuation, attained policy floors, and same-state first-exit splice compare every lifetime history without solving the unrestricted value function. The degree-5/4 economy shows why this is more than a different presentation of an order argument: both displayed order-based high modules are unavailable there, while the strict barrier certificate succeeds. This is necessity within the paper's modular architecture, not a claim that no other proof could exist.

The scope is deliberately narrow. The model is deterministic, has one state, and studies a planner. Selection does not by itself imply a common destination, as the annual realization makes explicit. Uncertainty, multidimensional capital, moving target sets, strategic interaction, and automated certificate construction are natural next steps; none is needed for the results proved here.

DATA AND CODE AVAILABILITY

This is a theoretical paper and uses no confidential, proprietary, or third-party data. All numerical objects reported in the paper are generated by the included verification programs and machine-readable certificates. The accompanying replication package contains the ex-

act certificates, independent replay programs, analytical checks, expected-output records, and a master verifier with smoke, full, and negative-control modes.

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